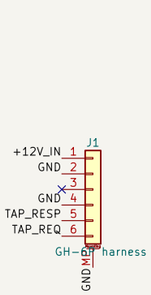
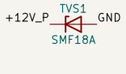
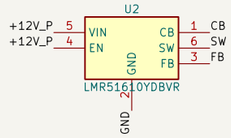
POWER: harness 12 V -> F1 PPTC -> D1 reverse-polarity -> TVS1/C1/C2 -> U2 buck 5.04 V -> D2 backfeed block -> C3/C4 -> XIAO 5V









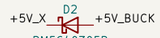










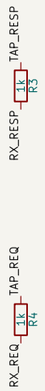






TAPS (PASSIVE ONLY): harness carries the off-board 10k series resistors; on-board R3/R4 1k are the second series element.

R5/R6 pull-ups and C5/C6 filters are DNP footprints. No default-populated part lets a XIAO pin drive a tap line.

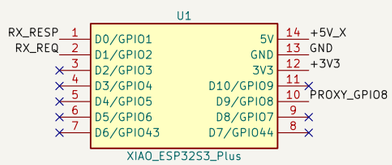












DNP: proxy only. R12 + JP1 (open) + TP1 reserve a future GPIO8 TX path; TP1 is NOT wired to J1.

Fitting R12 and bridging JP1 requires an explicitly approved proxy-mode change.







NO USB WHILE INSTALLED without a USB isolator (XIAO 5V is fed from the inverter through D2).

J1 pinout: 1 +12V, 2 GND (power), 3 spare (NC), 4 GND (signal), 5 TAP pad 9 (inverter replies), 6 TAP pad 10 (dongle requests).

Passive Modbus tap carrier for XIAO ESP32S3 Plus (generated by scripts/gen_sch.py)

SolArk_Live_Data

Sheet: /

File: carrier.kicad_sch

Title: SolArk XIAO tap carrier

Size: A3	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1